



Measuring the charge of an electron from the Millikan oil drop experiment

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(Received January 23, 2024, Published online February 6, 2024)

The charge of an electron is a fundamental constant important in many fields across physics. Using the Millikan oil drop experiment, the charges of oil drops are determined based on the velocity at which each drop falls. By analyzing the charge of the drops due to excess electrons, all of the charges are determined to be approximate multiples of each other and are all separated by the charge of an electron. Once the charge of each oil drop is determined, it is compared to the predicted amount of excess electrons, allowing the electron charge to be approximated: $1.52 \pm 0.09 \times 10^{-19}$ C.

I. INTRODUCTION

The charge of an electron is considered fundamental in many fields of physics. Equations and properties of other particles have been developed under the assumption that the charge of electrons is accurate. Consequently, determining this charge accurately is essential.

The first attempt to determine the charge of an electron was conducted in the 1890's by Townsend [1]. His method included finding the charge and weight of an ionized gas cloud to calculate the electron charge at about 1×10^{-19} C. Similar experiments were conducted by two of Townsend's colleagues, J.J. Thompson and H.S. Wilson. However, in 1909 Robert A. Millikan built an updated design of the experiment with a higher voltage and calculated the electron charge value accepted until 1928, when the modern value was discovered through X-ray diffraction [2].

Using an apparatus similar to Millikan's, the velocity at which oil drops rise and fall is measured. This information is calculated to give the number of excess electrons on multiple drops of oil vapor. By analyzing the gaps between the charges on the oil drops, the charge of an electron is determined.

II. QUANTIFYING EXCESS ELECTRONS

The Millikan oil drop experiment uses electrostatic forces to control the velocity at which ionized oil drops fall. The apparatus is constructed so that drops of oil vapor fall within the confines of a capacitor. A voltage across the capacitor can be modified and reversed to change whether the drops rise or fall. An oil drop with a large number of excess electrons will move more quickly due to stronger attraction and repulsion forces. Consequently, the velocity of a drop is related to the amount of charge on that drop of oil. The charge can be calculated from the magnitude of the force the drop experiences in an electric field.

The forces acting on each oil drop include the electric force provided by the capacitor, gravity, and, to a small extent, the linear component of air resistance. The capacitor can be adjusted, causing the oil drops to rise and fall, allowing the system to be examined as a static system. If m is mass of the drop, g is acceleration due to gravity, k is an air resistance constant, v_f and v_r are the velocities of the drop as it falls and rises, E is the magnitude of the electric field, and q is the charge of the drop, then the system can be described as:

$$\begin{aligned} mg &= kv_f, \\ Eq &= mg + kv_r. \end{aligned} \quad (1)$$

The mass m is eliminated by assuming the oil drop is a sphere and using the equation for the volume of a sphere:

$$m = \frac{4}{3}\pi r^3 \rho, \quad (2)$$

where r and ρ are the drop's radius and density, respectively.

Stokes' Law replaces the drop's radius with velocity, as it relates the two for objects in a viscous medium. A correctional term is also added for the drop's low velocity:

$$r = \sqrt{\left(\frac{b}{2p}\right)^2 + \frac{9\eta v_f}{2g\rho}} - \frac{b}{2p}, \quad (3)$$

where p is the barometric pressure, η is the viscosity of air, and b is a constant.

These equations are combined and solved for the charge of the oil drop q in terms of its velocity v :

$$q = \frac{4}{3}\pi\rho g \left[\sqrt{\left(\frac{b}{2p}\right)^2 + \frac{9\eta v_f}{2g\rho}} - \frac{b}{2p} \right]^3 \frac{v_f + v_r}{Ev_f}. \quad (4)$$

Given eq. (4), the charge of each drop of oil can be determined by measuring the velocity of each drop. Having the total charge of each oil drop is crucial for calculating the charge of an electron.

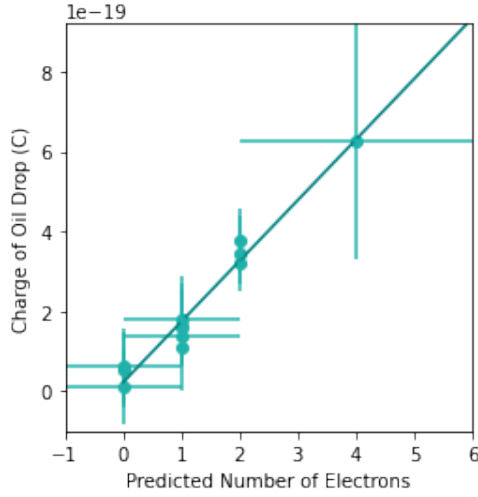


FIG. 1. The charge of each oil drop is plotted over the predicted number of electrons on each drop. The slope gives the experimental charge of an electron.

In the experiment, many trials were conducted to measure the velocity of different oil drops. A stop watch was used to measure the time it took for the drop to move a certain distance. Then, the drop velocity was calculated by dividing the distance covered by the time the drop took to move that distance.

The charges of each drop are then calculated from their respective velocities using eq. (4). Many of the charges are multiples of an integer, so, theoretically, the integer is the charge of an electron. The charge of each drop of oil is an integer multiple of the electron charge due to the sum of all the excess electrons on each drop, causing them to be ionized.

Once the charge of each drop is calculated, the different charges are grouped together based on the similarity of magnitude, as shown in table I. Based on the average difference between these groupings, the number of electrons can be predicted.

Then, to determine the charge of an electron, the charge of each oil drop is plotted over the predicted number of electrons in table fig. 1. The slope of the plot gives the experimental value of the charge of an electron.

From the slope, the electron charge is experimentally determined to be $1.52 \pm 0.09 \times 10^{-19}$ C. While this calcu-

Estimated Number of Electrons	Approximate Charge of Oil Drop [$\times 10^{-19}$ C]
0	0.0
1	1.6
2	3.2
4	6.4

TABLE I. The groupings of oil drop charges and the corresponding predicted number of electrons.

lation is not completely accurate, more trials and a timing device more precise than a stopwatch would improve this value. This calculation replicates that originally conducted by Millikan himself to determine the charge of an electron.

III. CONCLUSION

The Millikan oil drop experiment presents a method for determining the amount of charge on drops of oil vapor, and from here, the charge of an electron. In conducting this experiment, the charge of multiple oil drops was determined and plotted. From this data, the charge of an electron was approximated to be $1.52 \pm 0.09 \times 10^{-19}$ C by determining the amount of excess electrons on each drop of oil.

Future work for this experiment would be refining the acquired measurements. Many of the uncertainty ranges in this experiment are larger than ideal and more precision in timing the oil drops would greatly improve this data. Determining the charge of the electron was originally work completed by Robert A. Millikan and is significant in its relevance throughout physics.

Appendix A: Uncertainty analysis

The uncertainty of most of the data was based on the uncertainty of the velocity of each drop. This was found by dividing the range of velocities for each drop by two. For the drops that only had one trial, the uncertainty was approximated as the average uncertainty of the drops with multiple trials. Once the velocity uncertainty was determined, it was scaled according to the equations to find the charge uncertainty, similar to how the charge was derived from velocity. This is the uncertainty that appears in fig. 1 for the charge of each oil drop.

The uncertainty in the number of predicted electrons is found by comparing the magnitude of the charge uncertainty to the calculated electron charge. This is then converted to an integer as the excess number of electrons on each drop has to be an integer.

To determine the uncertainty in the charge of the electron, the uncertainty in the slope of fig. 1 was determined through the use of a covariance matrix in python.

Variable	Value	Units
Slope	1.52 ± 0.09	10^{-19} C
Intercept	2 ± 2	$\times 10^{-20}$ C

TABLE II. The slope and y-intercept values of fig. 1.

ACKNOWLEDGEMENTS

I appreciate my collaboration with Sydney Pelnar, my lab partner; Justin Wheeler for advice on data collection; and Carissa Kiehl for advice on data analysis.

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- [1] S. Bishop, Millikan oil drop apparatus.
[2] T. E. of Encyclopaedia, Britannica (2022).